



MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Thermal Properties & Temperature

2.2

STUDENT NAME



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PHYSICS WITH
MOIZ DAWOODI

SYLLABUS OUTLINE:

2.2 Thermal properties and temperature**2.2.1 Thermal expansion of solids, liquids and gases**

- 1 Explain applications and consequences of thermal expansion in the context of common examples, including the liquid-in-glass thermometer
- 2 Explain, in terms of the motion and arrangement of particles, the thermal expansion of solids, liquids and gases, and state the relative order of magnitudes of the expansion of solids, liquids and gases
- 3 Convert temperatures between kelvin and degrees Celsius; recall and use the equation

$$T \text{ (in K)} = \theta \text{ (in } ^\circ\text{C)} + 273$$

2.2.2 Specific heat capacity

- 1 Know that an increase in the temperature of an object increases its internal energy
- 2 Describe an increase in temperature of an object in terms of an increase in the average kinetic energies of all of the particles in the object
- 3 Define specific heat capacity as the energy required per unit mass per unit temperature increase; recall and use the equation

$$\text{specific heat capacity} = \frac{\text{change in energy}}{\text{mass} \times \text{change in temperature}}$$

$$c = \frac{\Delta E}{m\Delta\theta}$$

- 4 Describe experiments to measure the specific heat capacity of a solid and of a liquid

2.2.3 Melting, boiling and evaporation

- 1 Describe melting, solidification, boiling and condensation in terms of energy transfer without a change in temperature
- 2 Know the melting and boiling temperatures for water at standard atmospheric pressure
- 3 Describe the differences between boiling and evaporation
- 4 Describe evaporation in terms of the escape of more energetic particles from the surface of a liquid
- 5 Describe how temperature, surface area and air movement over a surface affect evaporation
- 6 Explain how evaporation causes cooling
- 7 Describe latent heat as the energy required to change the state of a substance and explain it in terms of particle behaviour and the forces between particles

THERMAL EXPANSION:

"Expansion of matter due to heat is called thermal expansion."

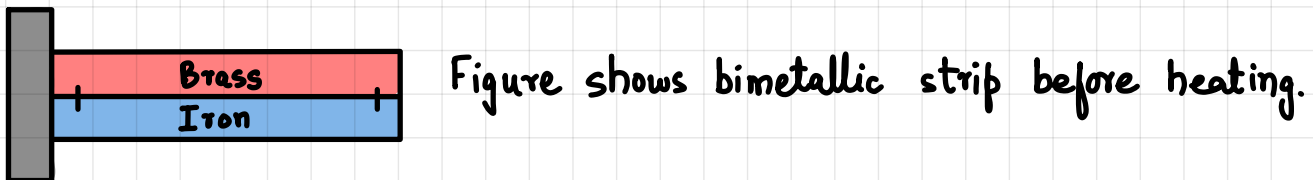
- On heating matter expands & on cooling, matter contracts.
- Thermal expansion is highest in gases as the particles are far away from each other & have less intermolecular forces.
- Thermal expansion is lowest in solids as the particles are very closer with each other & have strong intermolecular forces.

USES OF THERMAL EXPANSION:

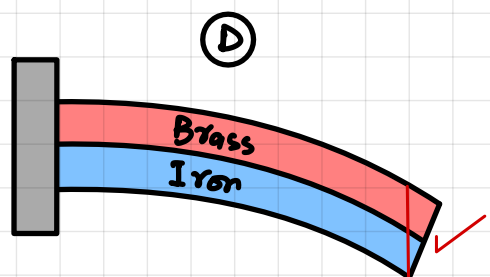
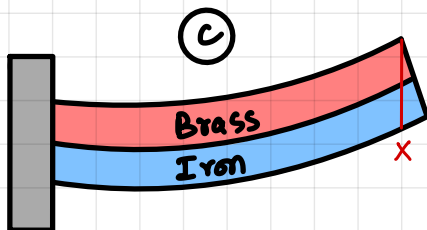
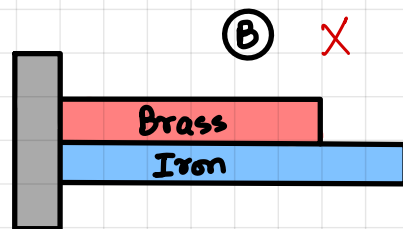
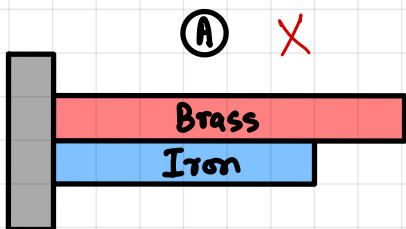
Brass expands more than iron.

BIMETALLIC STRIP:

Bimetallic strip is made up of two different metals riveted together

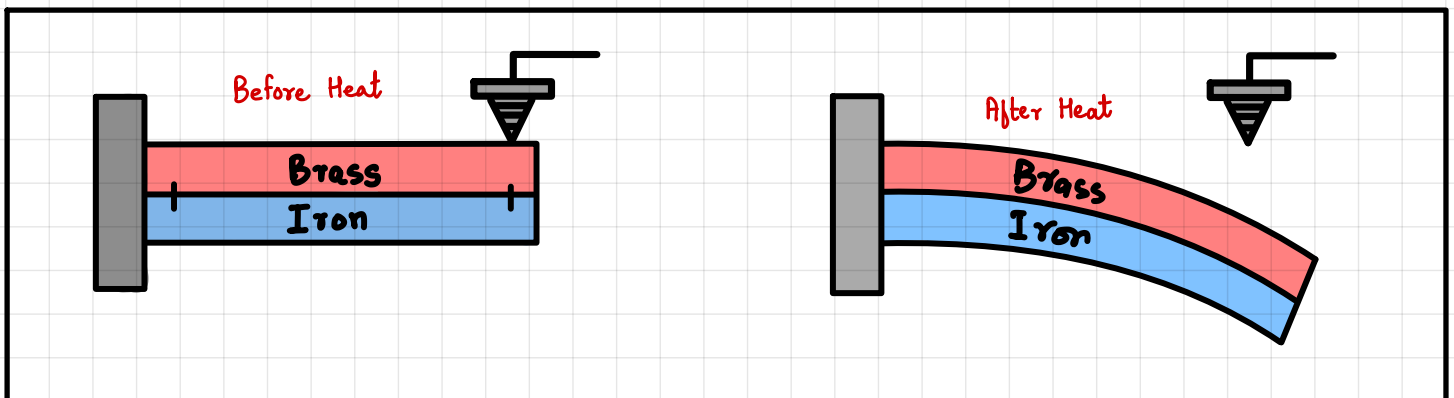


Which diagram is correct when bimetallic strip is heated?



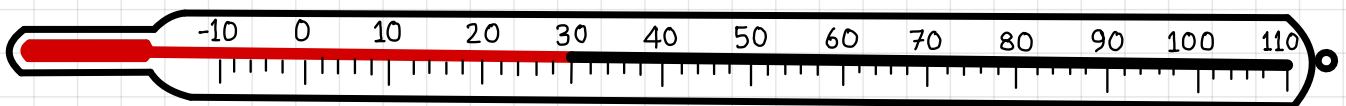
• BIMETALLIC THERMOSTAT:

"A device which is used to keep the temperature constant using bimetallic strips is called bimetallic thermostat.



• LIQUID-IN-GLASS THERMOMETER:

A liquid in glass thermometer contains mercury inside bulb which contracts upon cooling & expands upon heating.



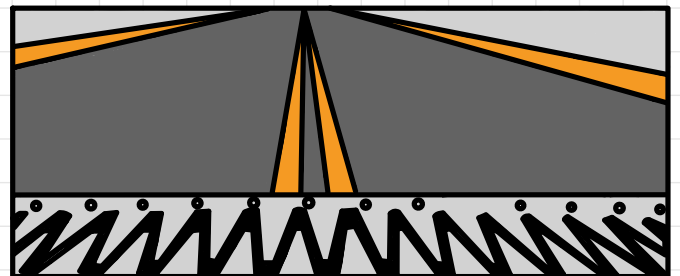
CONVERSION OF DEGREE CELCIUS TO KELVIN:

$$\square ^\circ\text{C} \rightarrow \square + 273 \text{ K}$$

CONSEQUENCES OF THERMAL EXPANSION:

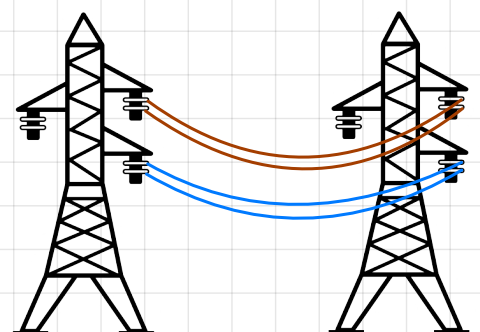
• EXPANSION JOINTS:

Expansion joints in bridges gives space for bridge to expand during summers.



• ELECTRIC CABLES:

Electric cables are suspended loosely so the cable has room for contraction in cold weather otherwise cable may break.



TEMPERATURE:

"The degree of hotness or coldness is called temperature."

When temperature increases, internal energy of the system also increases hence increase in average kinetic energy occurs.

SPECIFIC HEAT CAPACITY:

"Amount of heat required to change the temperature of unit mass by 1°C is called specific heat capacity."

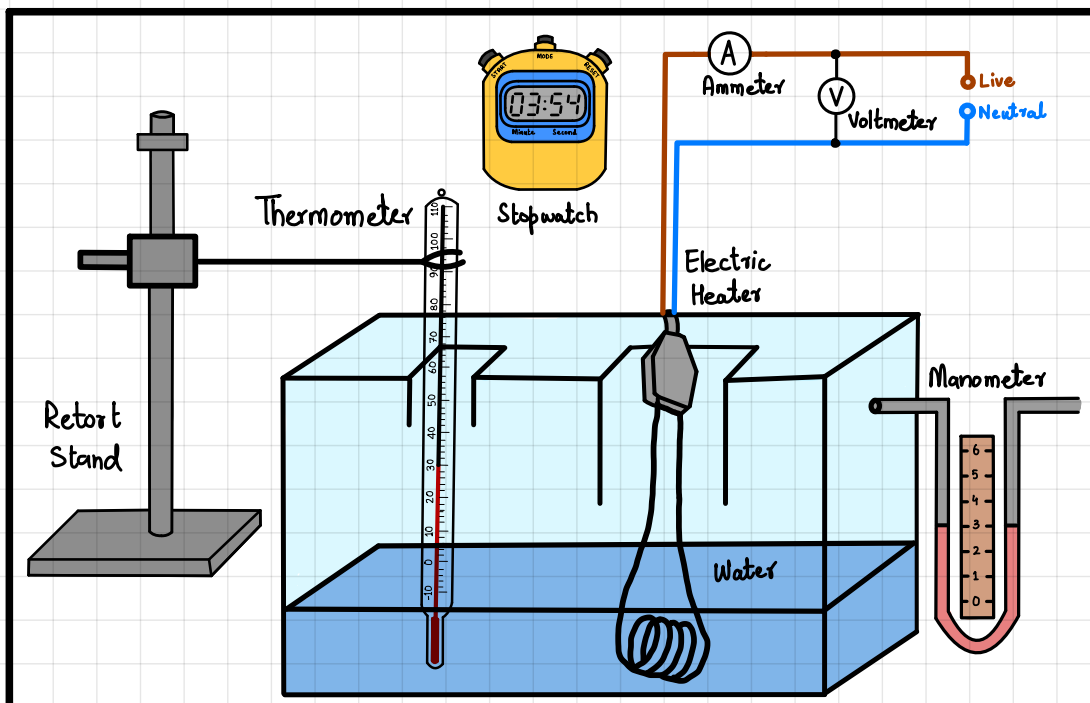
$$Q = mc \Delta T$$

FORMULA:

$$\text{Specific Heat Capacity} = \frac{\text{Amount of Heat}}{\text{Mass} \times \text{Change in Temperature}}$$

$$c = \frac{Q}{m \Delta T}$$

Its S.I unit is J/kg°C.

EXPERIMENT TO MEASURE SPECIFIC HEAT CAPACITY OF LIQUID:

Set up the apparatus as shown & follow steps.

All the electrical energy is converted into heat energy.

- Measure the mass of empty container 'M₁'.
- Measure the mass of container filled with water 'M₂'.
- Subtract both to get mass of water.
- First measure the initial temperature of water using thermometer & label this as 'T₁'.
- Switch on heater at the same time switch on stop watch.
- Measure the ammeter & Voltmeter reading as well.
- Now measure the final temperature as 'T₂' after 5 minutes (300 s).
- Subtract 'T₂ - T₁' to get temperature change.
- Put all the measurements in the formula to calculate specific heat.

$$c = \frac{Q}{m \Delta T} = \frac{P t}{m \Delta T}$$

$$c = \frac{IVt}{m \Delta T}$$

2.2.3

Melting, Boiling & Evaporation

SPECIFIC LATENT HEAT:

"Amount of heat required to change the state of unit mass without change in temperature is called specific latent heat."

Temperature remains constant because during state change, all the heat is used to make/break bonds.

FORMULA:

$$\text{Specific Latent Heat} = \frac{\text{Heat}}{\text{Mass}}$$

$$l = \frac{Q}{m}$$

Its S.I unit is J/kg.

Solid $\xrightleftharpoons{\text{Fusion}}$ Liquid $\xrightleftharpoons{\text{Vapourisation}}$ Gas



SPECIFIC LATENT HEAT OF FUSION:

"Amount of heat required to change the state of unit mass from solid to liquid or vice versa without change in temperature."

SPECIFIC LATENT HEAT OF VAPORIZATION:

"Amount of heat required to change the state of unit mass from liquid to gas or vice versa without change in temperature."

MELTING POINT:

"The specific temperature at which solid starts to convert into liquid is called melting point."

E.g: Melting point of ice is 0°C .

If a substance is pure, melting & freezing occurs at same temperature.

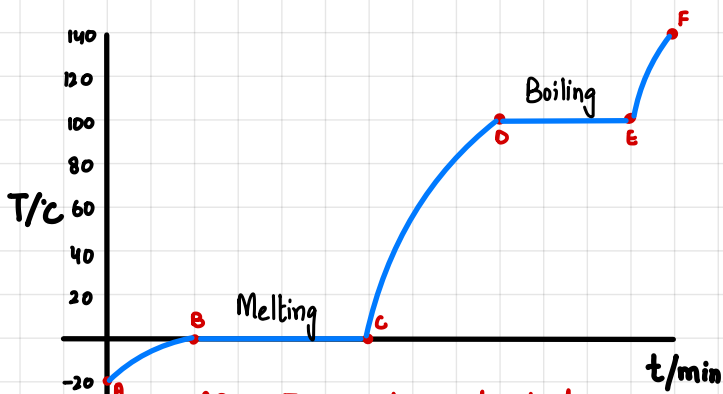
BOILING POINT:

"The specific temperature at which liquid starts to convert into gas is called boiling point."

E.g: Boiling point of water is 100°C .

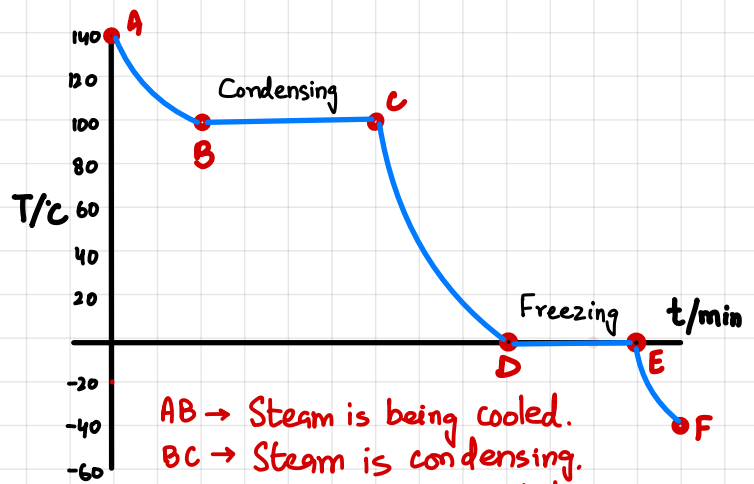
If a substance is pure, boiling & condensing occurs at same temperature.

HEATING CURVE:



AB → Ice is being heated.
 BC → Ice is melting.
 CD → Water is being heated.
 DE → Water is boiling.
 EF → Steam is heated

COOLING CURVE:



AB → Steam is being cooled.
 BC → Steam is condensing.
 CD → Water is being cooled.
 DE → Water is freezing.
 EF → Ice is cooled.

BOILING

- It happens with external heat.
- Bubbling occurs.
- Boiling occurs throughout the liquid.
- Boiling is a fast process.
- Boiling occurs at specific temperature.
- Heating Process.

EVAPORATION

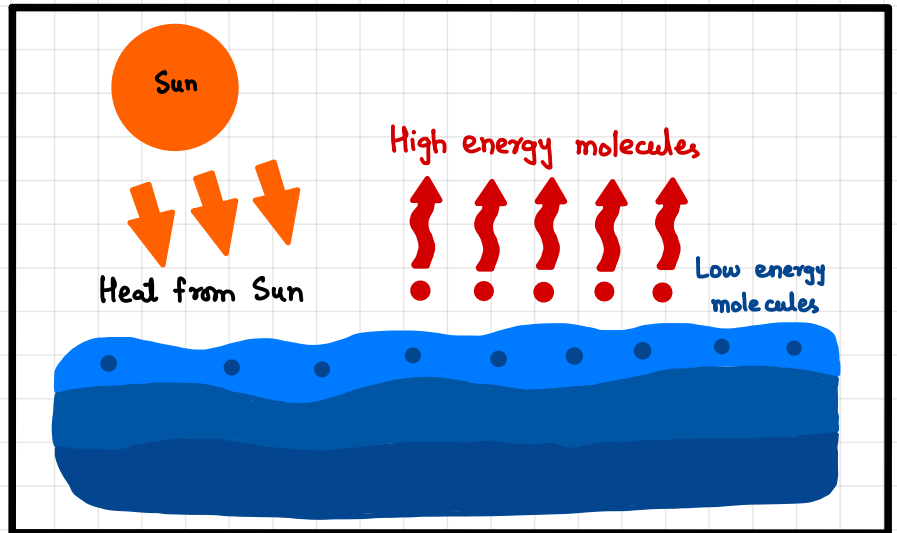
- It happens with surrounding heat.
- Bubbling doesn't occur.
- Evaporation occurs at the surface of liquid.
- Evaporation is a slow process.
- Evaporation occurs at any temperature.
- Cooling Process.

EVAPORATION.

"A change of state from liquid to gas which occurs at any temperature is called evaporation."

Evaporation only occurs at the surface of liquid.

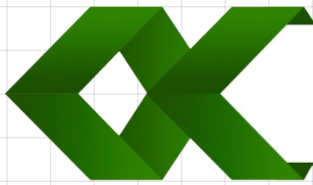
- Molecules at the surface of water receives heat, becomes high energetic, leaves the surface of liquid.



- This leaves lesser energetic (cold) molecules behind.
- So evaporation is a cooling process

FACTORS EFFECTING RATE OF EVAPORATION

SURFACE AREA	Evaporation increase when surface area is increased.
TEMPERATURE DIFFERENCE	Evaporation increase when temperature difference is increased.
WIND SPEED	Evaporation increase when wind/draught speed is increased.
DRYNESS OF AIR	Evaporation increase when dryness of air is increased.
NATURE OF LIQUID	Liquids having low boiling point have high evaporation rate.



MATCH THE FOLLOWING

Specific heat is the heat	required to change state of a body.
Latent heat is the heat	during state change.
Temperature remains same	required to change temperature of a body by 1°C .
Boiling is a fast process	of high energy leaves the surface of liquid.
In evaporation, molecules	than evaporation.
Evaporation gives	100°C & freezes at 0°C .
Water boils at	cooling effect.
Latent heat of vaporization	with greater surface area.
Evaporation increases	deals with state changes from liquid to gas or vice versa.
$\frac{\text{J}}{\text{kg}^{\circ}\text{C}}$ is the S.I Unit	degree of hotness or coldness.
Temperature is the	of specific heat capacity.

- 01 A balloon filled with air is gently heated.

What happens to the mass and the density of the air inside the balloon?

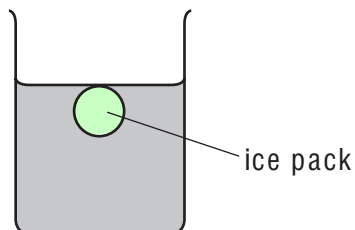
	mass	density
A	decreases	decreases ✓
B	decreases	stays the same
C	stays the same ✓	decreases ✓
D	stays the same ✓	stays the same

$$\downarrow \rho = \frac{m}{\uparrow V}$$

[MJ2011/P11/Q16]

- 02 An ice pack is used to cool 0.25 kg of water. The specific heat capacity of water is 4.2 kJ/(kg °C).

$$\begin{aligned} Q &= mc\Delta T \\ &= 0.25 \times 4.2 \times 15 \\ &= 15.75 \text{ kJ} \end{aligned}$$



How much thermal energy (heat) must the ice pack extract from the water to reduce the water temperature by 15 °C?

A 0.070 kJ

B 1.1 kJ

C 16 kJ

D 250 kJ

[MJ2011/P11/Q18]

- 03 Fillings in teeth should be made from a material which

A does not expand when heated.

B expands by the same amount as the tooth when heated.

C expands less than the tooth when heated.

D expands more than the tooth when heated.

[MJ2011/P11/Q19]

- 04 To raise the temperature of a 2.0 kg block of metal by 20 °C, energy of 5.2 kJ is needed.

What is the value of the specific heat capacity of the metal?

A 0.13 J/(kg °C)

B 52 J/(kg °C)

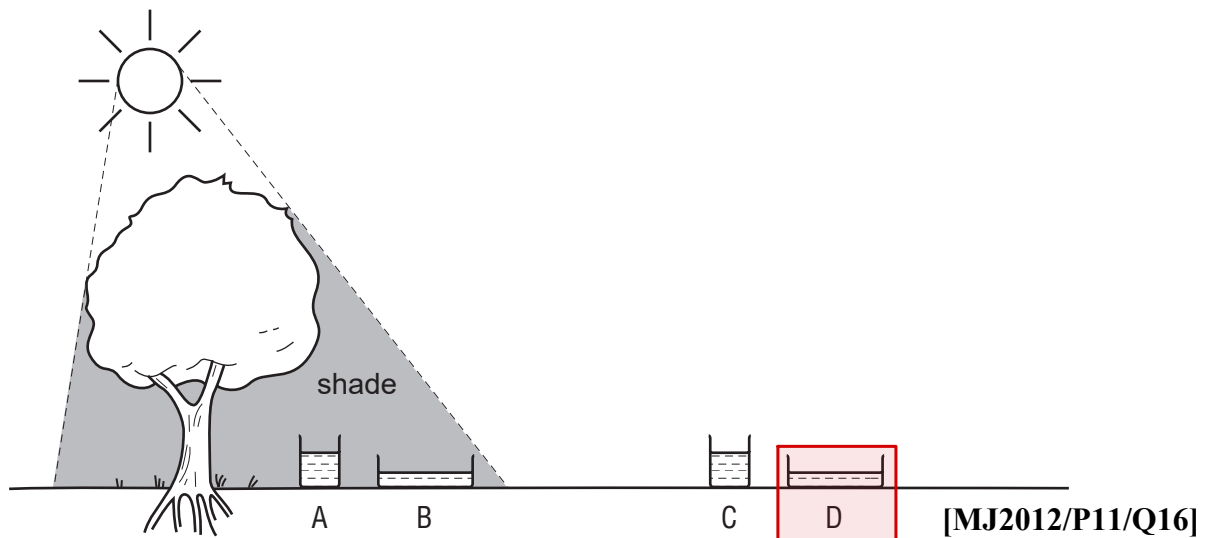
C 130 J/(kg °C)

D 52 000 J/(kg °C)

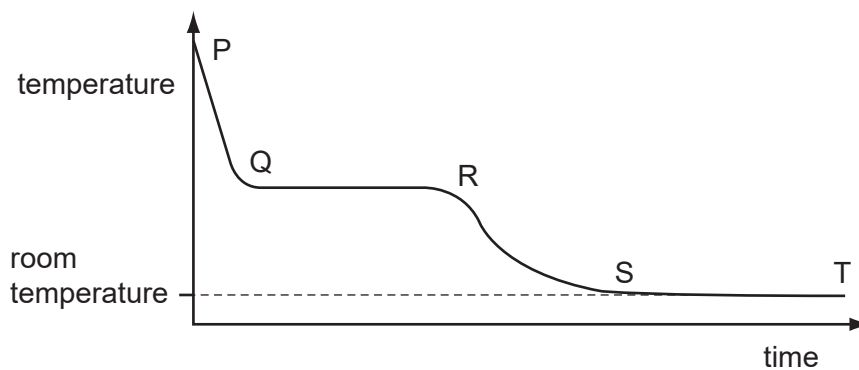
$$c = \frac{Q}{m\Delta T} = \frac{5.2 \times 1000}{2 \times 20}$$

[ON2011/P11/Q17]

- 05 Four containers are filled with equal volumes of water at the same temperature. Containers **A** and **B** are shaded by a tree. Sunlight falls on containers **C** and **D**. From which container does all the water evaporate first?



- 06 A hot liquid is allowed to cool. The graph shows the cooling curve.



In which part of the curve is latent heat released?

- A PQ **B QR** C RS D ST [MJ2012/P11/Q19]

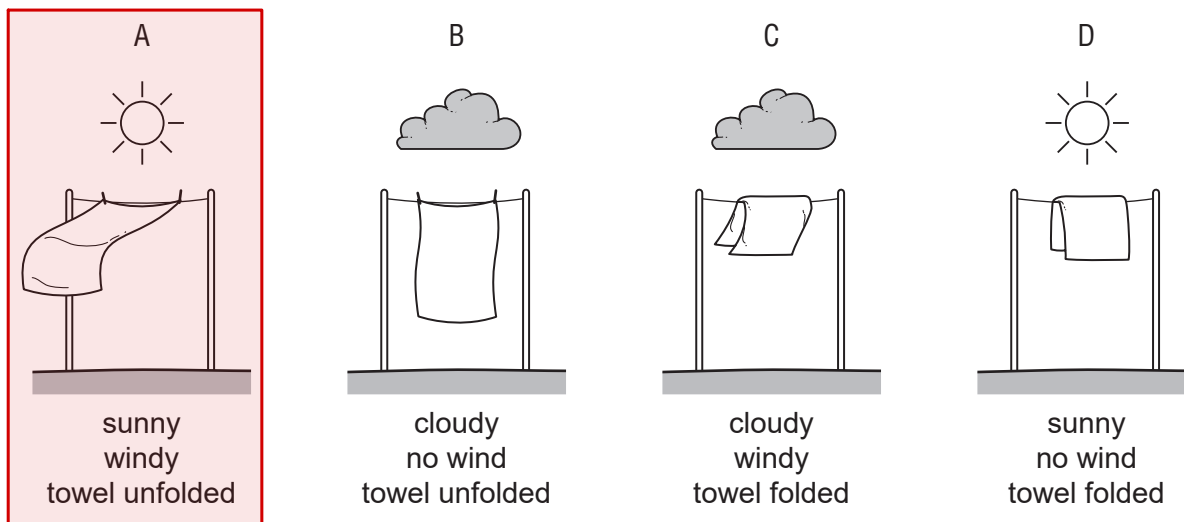
- 07 For the same temperature rise and the same original volume, which of the three states of matter expands the most and which state expands the least?

	expands most	expands least
A	gas ✓	solid ✓
B	liquid	gas
C	solid	gas
D	solid	liquid

[MJ2012/P11/Q20]

08 Four wet towels are hung out to dry as shown.

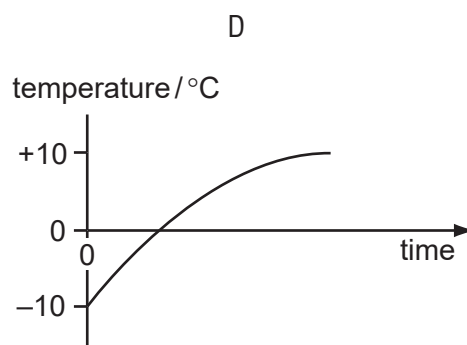
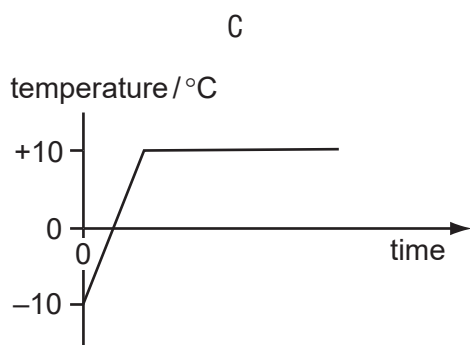
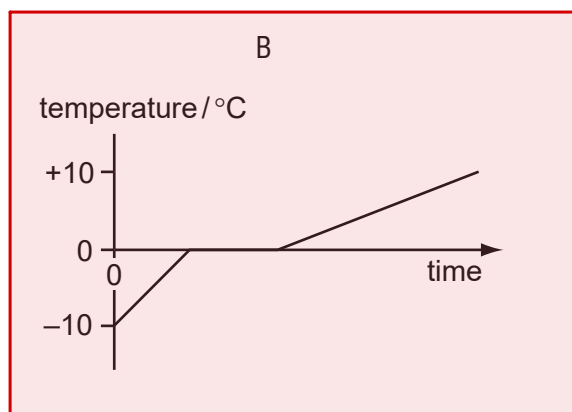
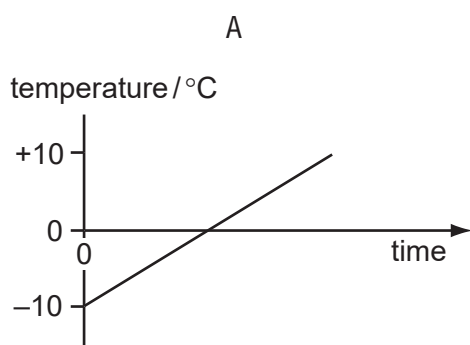
Which towel dries most quickly?



[MJ2012/P12/Q16]

09 Ice at -10°C is heated until it is water at $+10^{\circ}\text{C}$.

Which graph shows how the temperature changes with time?

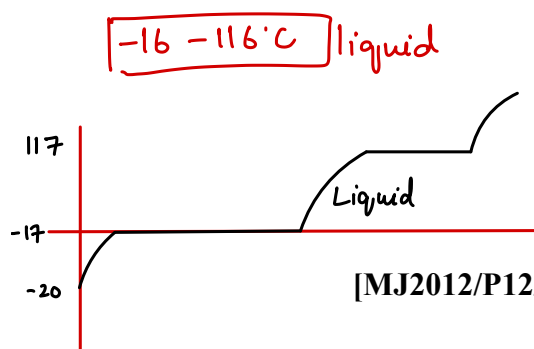


[MJ2012/P12/Q19]

10 A substance has a melting point of -17°C and a boiling point of 117°C .

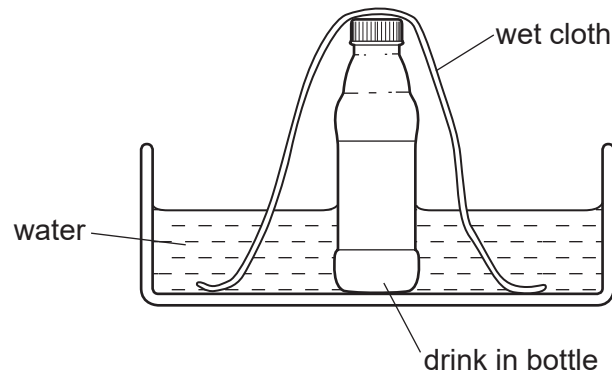
In which state does the substance exist at -10°C and at 110°C ?

	at -10°C	at 110°C
A	solid	liquid ✓
B	solid	gas
C	liquid ✓	liquid ✓
D	liquid ✓	gas



[MJ2012/P12/Q20]

- 11 On a hot day, the drink in a bottle can be kept cool by standing the bottle in a bowl of water and placing a wet cloth over it.



Why is the drink kept cool?

- A Hot air cannot escape from the bottle.
- B The cloth conducts heat from the bottle into the water.
- C The drink cannot evaporate from the bottle.

D Water evaporating from the cloth cools the drink.

[ON2012/P11/Q16]

- 12 Less heat is needed to raise the temperature of 1 kg of copper by 1 °C than is needed to raise the temperature of 1 kg of water by 1 °C.

$$\downarrow c = \frac{Q}{m\Delta T}$$

Which statement explains this?

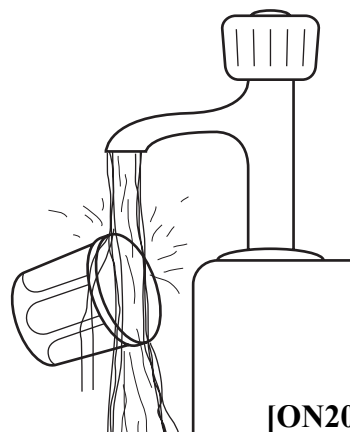
- A Copper has a higher melting point.
- B Copper has a lower specific heat capacity.**
- C Copper has a smaller specific latent heat.
- D Copper is a better conductor of heat.

[ON2012/P11/Q16]

- 13 A person cannot unscrew the metal lid of a pot of jam. The lid can be unscrewed after it has been held under hot, running water for a few seconds.

Why is this?

- A The air pressure in the jar falls.
- B The glass expands.
- C The jam melts.
- D The metal lid expands.**



[ON2012/P11/Q20]

- 14 In hot weather, people use electric fans to keep cool.

Why do the fans make them feel cool?

- A They change one form of energy into another.
- B They cool the air in the room.
- C They increase the rate of evaporation from the skin.
- D They speed up the vibration of air molecules.

[ON2012/P12/Q16]

- 15 An ice cube, at a temperature of 0°C , has a mass of 10 g. The specific latent heat of fusion of water is $3 \times 10^5 \text{ J/kg}$.

$$Q = ml = \frac{10}{1000} \times 3 \times 10^5$$

How much heat energy is needed to convert the ice cube into 10 g of water at 0°C ?

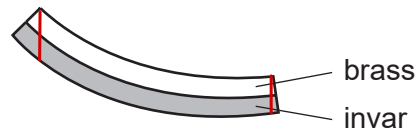
- A 30 J
- B 3000 J
- C $3 \times 10^4 \text{ J}$
- D $3 \times 10^6 \text{ J}$

[ON2012/P12/Q19]

- 16 A strip is made from two metals joined together. The diagrams show the strip at room temperature and after it has been cooled.



at room temperature



below room temperature

The change in shape occurs because

- A brass contracts more than invar.
- B brass expands when it cools down.
- C invar and brass contract by equal amounts.
- D invar contracts more than brass.

[ON2012/P12/Q20]

- 17 Using an electric kettle, 200 g of water at 100°C is converted into steam at 100°C in 300 seconds. The specific latent heat of steam is 2250 J/g .

What is the average electrical power used?

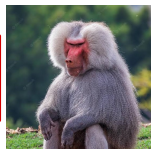
A $\frac{2250}{300 \times 200} \text{ W}$

B $\frac{200 \times 2250}{300} \text{ W}$

C $\frac{300 \times 2250}{200} \text{ W}$

D $200 \times 300 \times 2250 \text{ W}$

$$P = \frac{Q}{t} = \frac{ml}{t} = \frac{200 \times 2250}{300}$$



[MJ2013/P11/Q15]

18 Thermal energy of 12 000 J is supplied to a 2.0 kg mass of copper.

The specific heat capacity of copper is 400 J/(kg °C).

What is the rise in temperature?

- A** 15 °C **B** 30 °C **C** 60 °C **D** 100 °C

[MJ2013/P12/Q18]

19 What happens to the molecules of a gas when the gas changes into a liquid?

- A** They move closer and lose energy.
B They move closer and gain energy.
C They move apart and lose energy.
D They move apart and gain energy.

[MJ2013/P12/Q21]

20 When a person climbs out of a warm swimming pool on a hot summer day, he feels cold.

Why does this happen?

- A** The air has a high specific heat capacity.
B The air is a better conductor of heat than water.
C The water droplets increase his surface area.
D The water takes heat from his body to evaporate.

[ON2013/P11/Q17]

21 Different amounts of energy are supplied to copper blocks of different masses.

Which block experiences the greatest temperature change?

	mass of block / kg	energy supplied / J
A	0.1	200
B	0.2	200
C	0.4	600
D	0.8	400

[ON2013/P12/Q18]

22 The more energetic molecules in a liquid may escape from its top surface.

What is this process called?

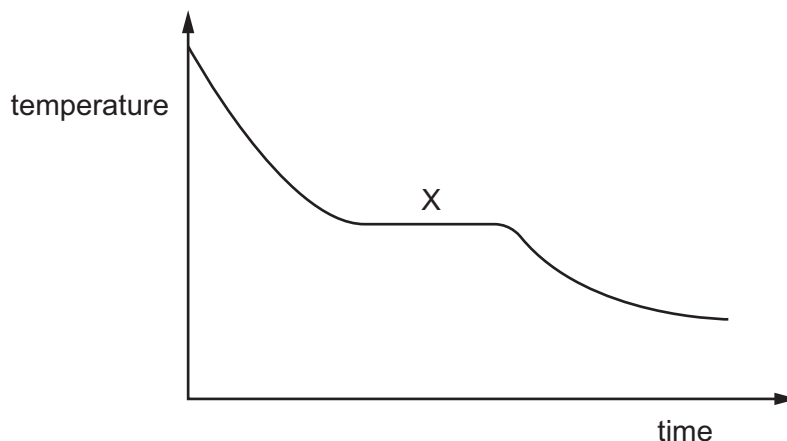
- A** boiling
B Brownian motion
C convection
D evaporation

[ON2013/P12/Q21]

- 23 A hot liquid is poured into a beaker. The graph shows how the temperature of the liquid changes as it cools towards room temperature.

What is occurring at region X?

- A boiling and evaporation
- B condensation only
- C evaporation only
- D solidification and evaporation



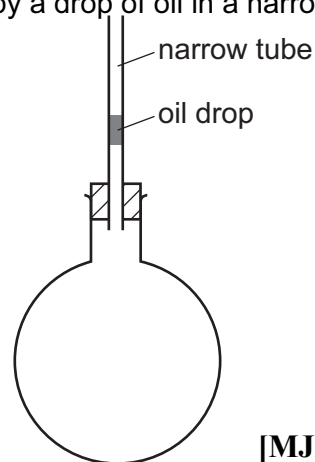
[MJ2014/P11/Q20]

- 24 The diagram shows a flask containing air. The air is trapped by a drop of oil in a narrow tube.

When the flask is heated the oil drop rises up the tube.

Which statement is **not** correct?

- A The air molecules each get larger.
- B The air molecules hit the container with greater force.
- C The air molecules move faster.
- D The air molecules move further apart.



[MJ2014/P12/Q21]

- 25 Thermal energy is transferred to a solid. First it melts and then it boils to produce a gas.

Which statement about the temperature is correct?

- A When melting and boiling the temperature does not change.
- B When melting and boiling the temperature increases.
- C When melting the temperature increases but when boiling the temperature stays the same.
- D When melting the temperature stays the same but when boiling the temperature increases.

[MJ2014/P12/Q22]

- 26 Steam at 100°C is passed into some water in a beaker. All the steam condenses in the water.

The mass of water in the beaker rises from 120.0 g to 122.0 g.

The specific latent heat of vaporisation of water is 2250 J/g.

How much thermal energy is lost by the steam as it condenses?

- A 8.9×10^{-4} J B 1.1×10^3 J C 4.5×10^3 J D 2.7×10^5 J [MJ2014/P12/Q23]

- 27 Water is poured into four dishes. In two of the dishes the water has a small surface area and in the other two it has a large surface area. The water in two of the dishes is cool and the water in the other two is warm.

From which dish does the water evaporate the quickest?

	surface area	temperature
A	large	cool
B	large	warm
C	small	cool
D	small	warm

[ON2014/P11/Q18]

- 28 Which substance in the table is liquid at 20 °C?

	melting point / °C	boiling point / °C
A	−218	−183
B	−39	357
C	44	280
D	119	444

[MJ2015/P11/Q19]

- 29 In a liquid, some energetic molecules break free from the surface even when the liquid is too cold for bubbles to form.

What is the name of this process?

- A** boiling
- B** condensation
- C** convection
- D** evaporation

[MJ2015/P12/Q21]

- 30 A pupil adds 37 g of ice at 0 °C to 100 g of water at 30 °C. The final temperature of the water and melted ice is 0 °C. No heat is lost to, or gained from, the surroundings.

The specific heat capacity of water is 4.2 J/(g °C).

What is the specific latent heat of ice?

- A** 47 J/g **B** 341 J/g **C** 4700 J/g **D** 12 600 J/g [MJ2015/P12/Q22]

- 31 The heat capacity of an object, of mass 2.0 kg, is C . The energy needed to

- A** increase the temperature of the whole object by Δt is $C\Delta t$.
- B** increase the temperature of unit mass of the object by Δt is $C\Delta t$.
- C** melt the whole object is C .
- D** melt unit mass of the object is C .

[MJ2015/P12/Q23]

- 32 An electric shower takes in cold water at 17°C . The shower gives 6000 J of energy every second to the cold water and heats it to 37°C . The specific heat capacity of water is $4200\text{ J}/(\text{kg }^{\circ}\text{C})$.

What is the mass of hot water supplied by the shower in one second?

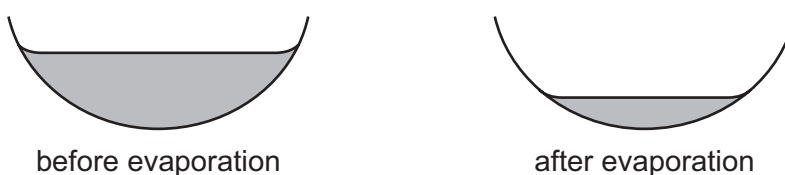
- A** 0.035 kg **B** 0.039 kg **C** 0.071 kg **D** 0.084 kg [ON2015/P11/Q17]

- 33 What is latent heat of vaporisation?

- A** the energy required to make molecules expand
B the energy required to make molecules expand and move apart
C the energy required to make molecules move apart
D the energy required to make molecules move faster

[ON2015/P11/Q18]

- 34 Some of the liquid in a dish evaporates, as shown in the diagrams.



Which molecules leave the liquid and which molecules in the liquid have greater average kinetic energy?

	molecules that leave have	molecules in the liquid have greater average kinetic energy
A	high energy	before evaporation
B	high energy	after evaporation
C	low energy	before evaporation
D	low energy	after evaporation

[ON2015/P12/Q18]

- 35 An insulated beaker contains 300 g of water, initially at 30°C . Water at 100°C is added until the temperature of the mixture reaches 50°C .

The specific heat capacity of water is $4.2\text{ J}/(\text{g }^{\circ}\text{C})$.

How much water is added?

- A** 60 g **B** 120 g **C** 180 g **D** 750 g [ON2015/P12/Q19]

ANSWER KEY				
1	C		21	A
2	C		22	D
3	B		23	D
4	C		24	A
5	D		25	A
6	B		26	C
7	A		27	B
8	A		28	B
9	B		29	D
10	C		30	B
11	D		31	A
12	B		32	C
13	D		33	C
14	C		34	A
15	B		35	B
16	A			
17	B			
18	A			
19	A			
20	D			

16°C → 0°C Specific Heat 0° Specific Latent Heat 0° - (-5°C) Specific Heat

- 1 Water at a temperature of 16°C enters an ice-making machine and emerges as ice cubes at a temperature of -5°C. The melting point of ice is 0°C.

- (a) Calculate the total energy removed from 1.0 kg of water as it cools from 16°C, changes into ice, and then cools to -5°C.

specific heat capacity of liquid water = $4.2 \times 10^3 \text{ J/(kg } ^\circ\text{C)}$

specific latent heat of fusion of water = $3.4 \times 10^5 \text{ J/kg}$

specific heat capacity of ice = $2.1 \times 10^3 \text{ J/(kg } ^\circ\text{C)}$

16 → 0°C

$$Q = mc\Delta T = 1 \times 4.2 \times 10^3 \times 16 = 67.2 \times 10^3 \text{ J}$$

0°C

$$Q = ml = 1 \times 3.4 \times 10^5 = 3.4 \times 10^5 \text{ J}$$

$$67.2 \times 10^3 + 3.4 \times 10^5 + 1.05 \times 10^4$$

0° → -5°C

$$Q = mc\Delta T = 1 \times 2.1 \times 10^3 \times 5 = 1.05 \times 10^4 \text{ J}$$

energy = $4.17 \times 10^5 \text{ J}$ [3]

- (b) Using ideas about molecules,

- (i) explain why energy is needed to change ice into water,

Energy is required to break bonds & move particles away from each other.

..... [1]

- (ii) suggest why less energy is needed to change ice into water than to change the same mass of water into steam.

$$\downarrow Q = ml \downarrow$$

Ice has less specific latent heat than water

..... [1]

[MJ2011/P21/Q4]

- 2 (a) In an experiment to measure the specific heat capacity of water, an electric heater heats water in a glass beaker. The temperature of the water is measured at regular intervals of time. Fig. 2.1 shows how the temperature varies with time t .

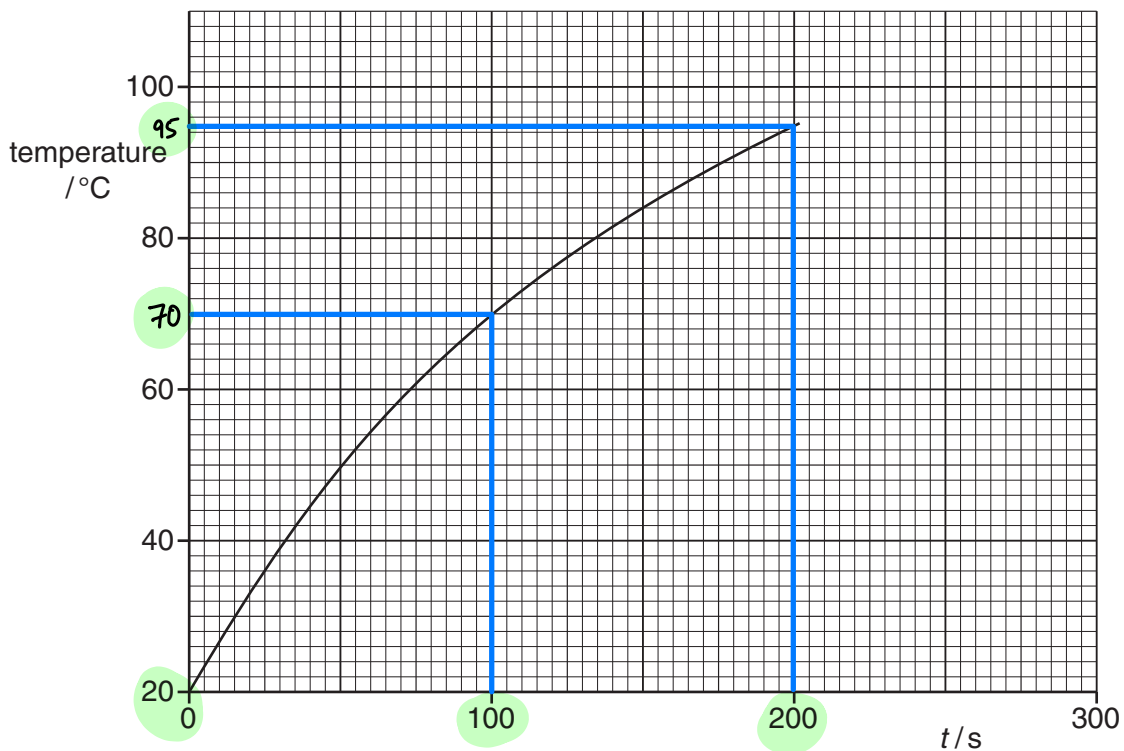


Fig. 2.1

- (i) Use Fig. 2.1 to determine the change in temperature between

$t = 0$ and $t = 100$ s,

$$70 - 20$$

change = 50°C

$t = 100$ s and $t = 200$ s.

$$95 - 70$$

change = 25°C

[1]

- (ii) State and explain why the values in (i) are different.

As temperature increases so heat loss due to evaporation increases so temperature of liquid doesn't rise as it should be.

[2]

- (iii) Describe and explain what happens to the water if the heating is continued.

Water will reach at boiling point & temperature stops rising for some time due to state change (mass of water will be less due to evaporation).

[2]

- (b) (i) The experiment in (a) is repeated using 72 g of water. The heater supplies 7400 J of thermal energy (heat) to the water and the temperature rise of the water is 23°C.

Calculate the specific heat capacity of water.

$$c = \frac{Q}{m\Delta T} = \frac{7400}{72 \times 23} = 4.46$$

- (ii) A bullet of mass $\frac{72}{1000} \rightarrow 72 \times 10^{-3} \text{ kg}$ is fired from a gun at a speed of 450 m/s. specific heat capacity = $4.5 \frac{\text{J}}{\text{g}^\circ\text{C}}$ [2]

Calculate the kinetic energy of the bullet.

$$\text{K.E} = \frac{1}{2}mv^2 = \frac{1}{2} \times 72 \times 10^{-3} \times (450)^2$$

$$\text{energy} = 7290 \text{ J} \text{[3]}$$

- (iii) The amount of internal energy gained by the water and the amount of kinetic energy gained by the bullet are approximately equal.

Describe the change in the motion of the molecules of the water and of the molecules of the bullet that this addition of energy has caused.

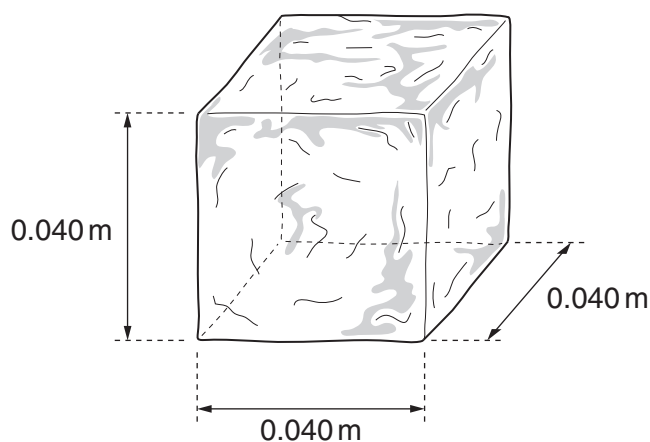
water: Molecules of water move vigorously randomly with high speeds.

bullet: Molecules of bullet vibrate faster carry energy from hot to cold end.

[3]

[MJ2011/P22/Q11]

3 Fig. 3.1 shows an ice cube at 0°C.



$V = l^3$ Fig. 3.1

The sides of the cube are of length 0.040 m. Ice at 0°C has a density of 920 kg/m³.

(a) Calculate

(i) the mass of the ice cube,

$$\rho = \frac{m}{V}$$

$$m = \rho V = 920 (0.040)^3 = 0.0588 \text{ kg}$$

$$\text{mass} = \dots 0.0588 \text{ kg} \approx 0.06 \text{ kg} \dots [3]$$

(ii) the weight of the ice cube.

$$W = mg = 0.06 \times 10$$

$$\text{weight} = \dots 0.6 \text{ N} \dots [1]$$

(b) The specific latent heat of fusion of ice is $3.4 \times 10^5 \text{ J/kg}$. Calculate the thermal energy (heat) absorbed by the ice cube as it melts.

$$Q = ml = 0.06 \times 3.4 \times 10^5$$

$$\text{thermal energy} = \dots 20400 \text{ J} \dots [2]$$

[ON2011/P21/Q1]

- 4 A lead bullet of mass 1.9 g is fired from a rifle in a sports club. The bullet misses the target and embeds itself in a wall behind the target. The bullet melts as it is stopped by the wall. The specific latent heat of fusion of lead is $2.2 \times 10^4 \text{ J/kg}$.

(a) State what is meant by *melting point*.

The specific temperature at which solids turns to liquid.

.....[1]

(b) (i) Calculate the energy required to melt the bullet, at its melting point, without raising its temperature.

$$Q = ml = \frac{1.9}{1000} \times 2.2 \times 10^4$$

energy = 41.8 J[3]

(ii) Assume that the energy that melts the bullet is equal to its kinetic energy just before it strikes the wall. Calculate the speed of the bullet just before it strikes the wall.

$$Q = K.E$$

$$Q = \frac{1}{2} mv^2$$

$$\sqrt{v^2} = \sqrt{\frac{2Q}{m}}$$

$$v = \sqrt{\frac{2 \times 41.8}{1.9 \times 10^{-3}}} = 209.76 \frac{\text{m}}{\text{s}}$$

speed = 210 $\frac{\text{m}}{\text{s}}$ [3]

(iii) Suggest two reasons why the speed of the bullet as it leaves the rifle is greater than the value calculated in (ii).

1. Some energy is lost as heat due to air resistance.

2. Some energy is lost as sound.

[2]

(c) Describe how the molecular structure of the lead changes as it melts.

Particles move away from each other & starts moving past each other with high speed & fairly strong forces.

.....

.....

.....[3]

- (d) On another occasion, lead bullets of twice the mass are used. One of these heavier bullets hits the wall with the speed calculated in (b)(ii). State and explain whether this bullet melts as it is stopped by the wall.

As mass is increased so more heat energy is required to melt the heavier bullet as compared to b(ii). Bullet at this energy will not melt

[3]

[ON2011/P22/Q10]

- 5 A student slowly heats a sample of solid wax in a test-tube. Fig. 5.1 shows how the temperature of the wax varies with time t .

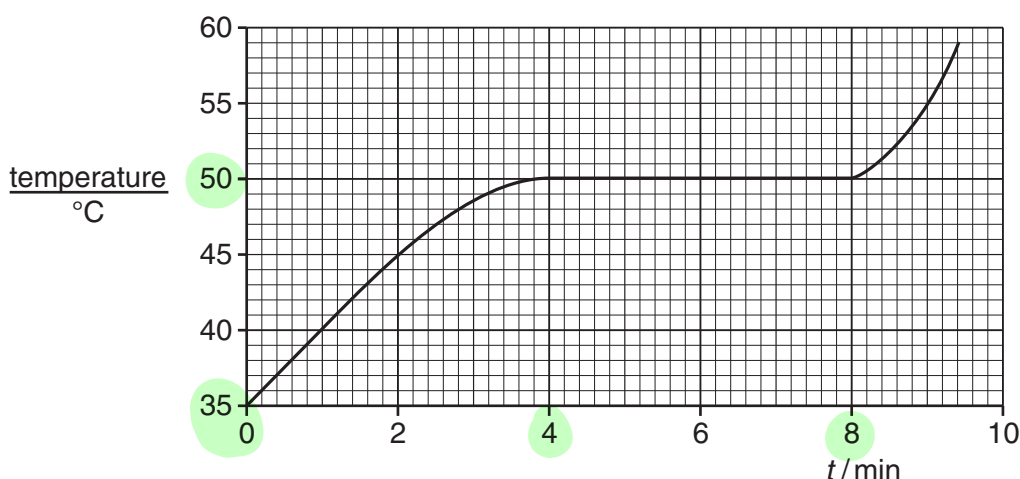


Fig. 5.1

- (a) State what is meant by the melting point of a substance.

The specific temperature at which solid turns to liquid.

.....[1]

- (b) Thermal energy (heat) passes into the wax throughout the experiment.

- (i) Describe what effect this energy has on the wax between $t = 0$ and $t = 4$ min.

Temperature of wax increases so the particles of wax vibrate vigorously.

.....[1]

- (ii) Describe and explain what effect this energy has on the wax between $t = 4$ min and $t = 8$ min.

Energy is utilised to break bonds & particles are moving away from each other as melting occurs.

.....[2]

- (c) The wax expands as the temperature rises. It expands more between $t = 8$ min and 9 min than between $t = 0$ and 1 min. Explain why.

Between 8-9 min, wax is liquid & liquid expands more than solid (0-1 min).

.....[1]

- 6 A beaker contains 60 g of a hot substance, initially in the liquid state. Fig. 6.1 shows how the temperature of the substance changes with time t as it cools in a laboratory.

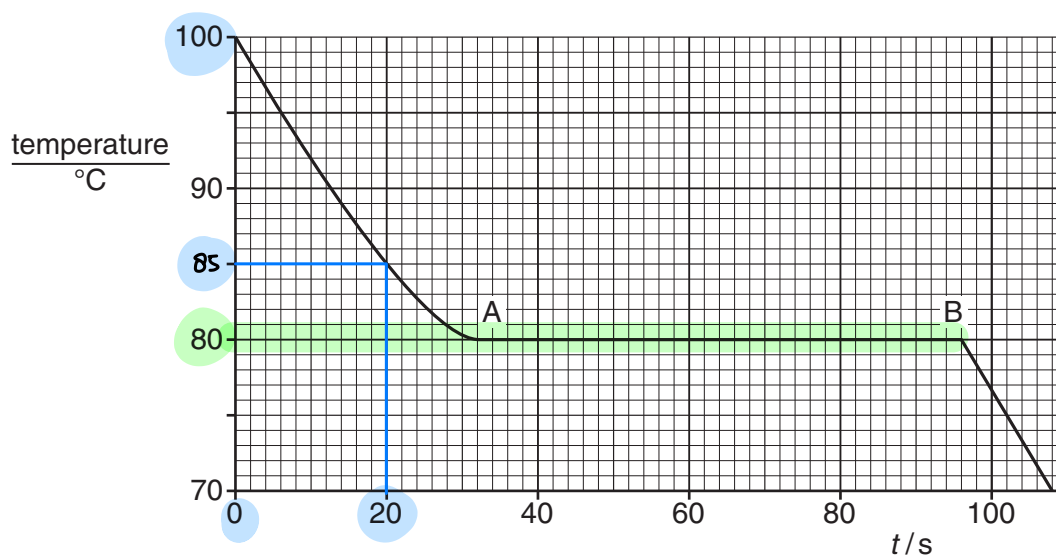


Fig. 6.1

- (a) Use Fig. 6.1 to determine the melting point of the substance.

80°C

[1]

- (b) The specific heat capacity of the liquid is $1.7\text{ J/(g }^{\circ}\text{C)}$. Calculate the loss of thermal energy (heat) from the liquid between $t = 0$ and $t = 20\text{ s}$.

$$Q = mc\Delta T = 60 \times 1.7 \times (100 - 85)$$

loss of thermal energy = 1530 J [2]

- (c) Between points A and B on Fig. 6.1, the temperature is constant as the substance changes from liquid to solid.

- (i) Explain why the temperature stays constant, even though thermal energy is lost by the substance.

All the heat is used to make bonds & none of the heat is left for thermometer. [2]

- (ii) Describe the change in the arrangement of the molecules as the substance changes from a liquid to a solid.

Particles move closer with each other & vibrate about their fixed positions [2]

- 7 Fig. 7.1 shows a domestic heater that is used to heat a room. It contains 16 large bricks each of mass 7.5 kg.

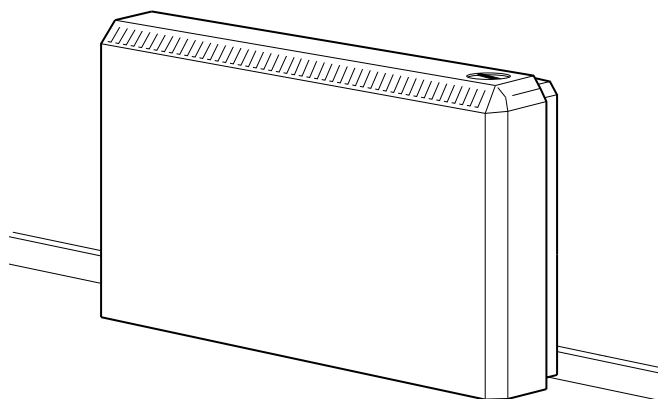


Fig. 7.1

During the night, the bricks are heated from a temperature of 17°C to 96°C . The bricks are made from a material that has a specific heat capacity of $2300\text{ J}/(\text{kg }^{\circ}\text{C})$.

- (a) Calculate the thermal energy (heat) supplied to the bricks.

thermal energy = [3]

- (b) During the day, the bricks gradually cool and the stored thermal energy is released to the room. After 7.0 hours, the bricks have cooled to 17°C .

- (i) Calculate the average rate of release of thermal energy to the room during these 7.0 hours.

rate = [2]

- (ii) At the beginning of the day, the heater releases thermal energy at a greater rate than later in the day.

Suggest why.

.....
.....
..... [2]

- (c) Fig. 7.2 is a diagram of the heater standing on the floor of the room and placed next to one of the walls.

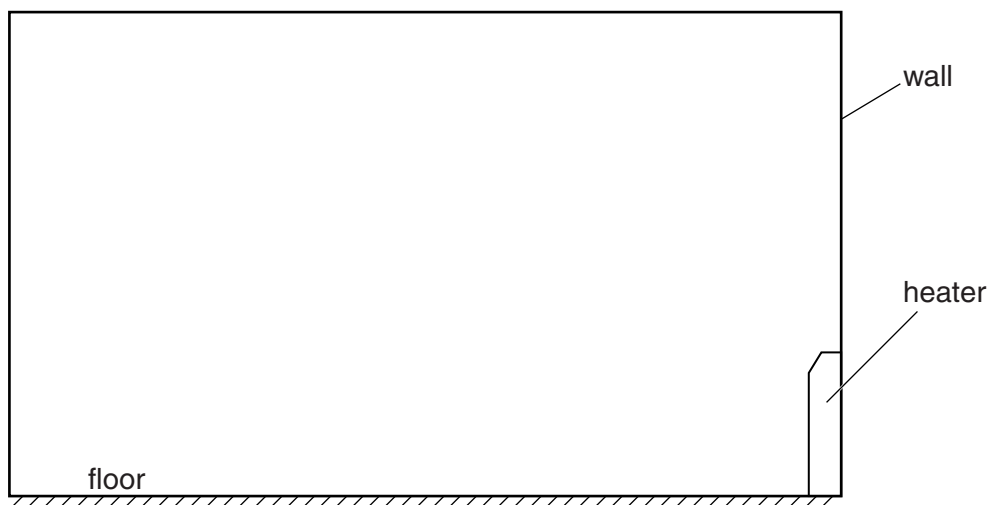


Fig. 7.2

Describe, and explain in detail, how the thermal energy is transferred throughout the room. You may draw on Fig. 7.2.

.....

.....

.....

.....

.....

..... [5]

- (d) State one method of thermal insulation that is used to keep a room warm, and explain why it is effective.

.....

.....

.....

.....

..... [3]

8 (a) The average input power to a freezer is 80 W. The cost of 1 kWh is 25 cents.

(i) Explain what is meant by the *kilowatt-hour* (kWh).

.....
.....[2]

(ii) Calculate the cost of running the freezer for one week.

cost =[3]

(b) A large jug containing 1.5 kg of water is placed in the freezer.

(i) The water cools from 25 °C to 0 °C in a time of 60 minutes. The specific heat capacity of water is 4.2 J/(g °C).

Calculate the thermal energy (heat) removed from the water as it cools from 25 °C to 0 °C.

energy removed =[3]

(ii) After the water has reached 0 °C, thermal energy is removed from the water at the same rate as in (i).

The specific latent heat of fusion of water is 3.3×10^5 J/kg.

Calculate the mass of water at 0 °C that becomes ice in 60 minutes.

mass =[2]

(iii) Describe the arrangement and the movement of the molecules

1. in liquid water,

.....
.....
.....
.....[2]

2. in ice.

.....
.....
.....
.....[2]

(iv) Ice at 0°C becomes water at 0°C.

State what, if anything, happens to the kinetic energy and the potential energy of the molecules as this happens.

kinetic energy:

potential energy:
[1]

[MJ2013/P21/Q10]

- 9** A swimming pool is filled with 450 m^3 of water at 12°C .

The density of water is 1000 kg/m^3 and its specific heat capacity is $4.2\text{ J/(g}^\circ\text{C)}$.

- (a)** Calculate the mass of water in the swimming pool.

mass =[2]

- (b) (i)** Calculate the thermal energy (heat) needed to raise the temperature of the water to 27°C .

thermal energy =[3]

- (ii)** Suggest one reason why the thermal energy supplied to the water by the heating system, when raising the temperature of the water to 27°C , differs from the value calculated in **(i)**.

.....

.....[1]

[ON2013/P21/Q1]

- 10** A large test-tube contains wax above its melting point. It is placed in a cool room.

Fig. 10.1 shows how the temperature T of the wax changes in a time of 30 minutes.

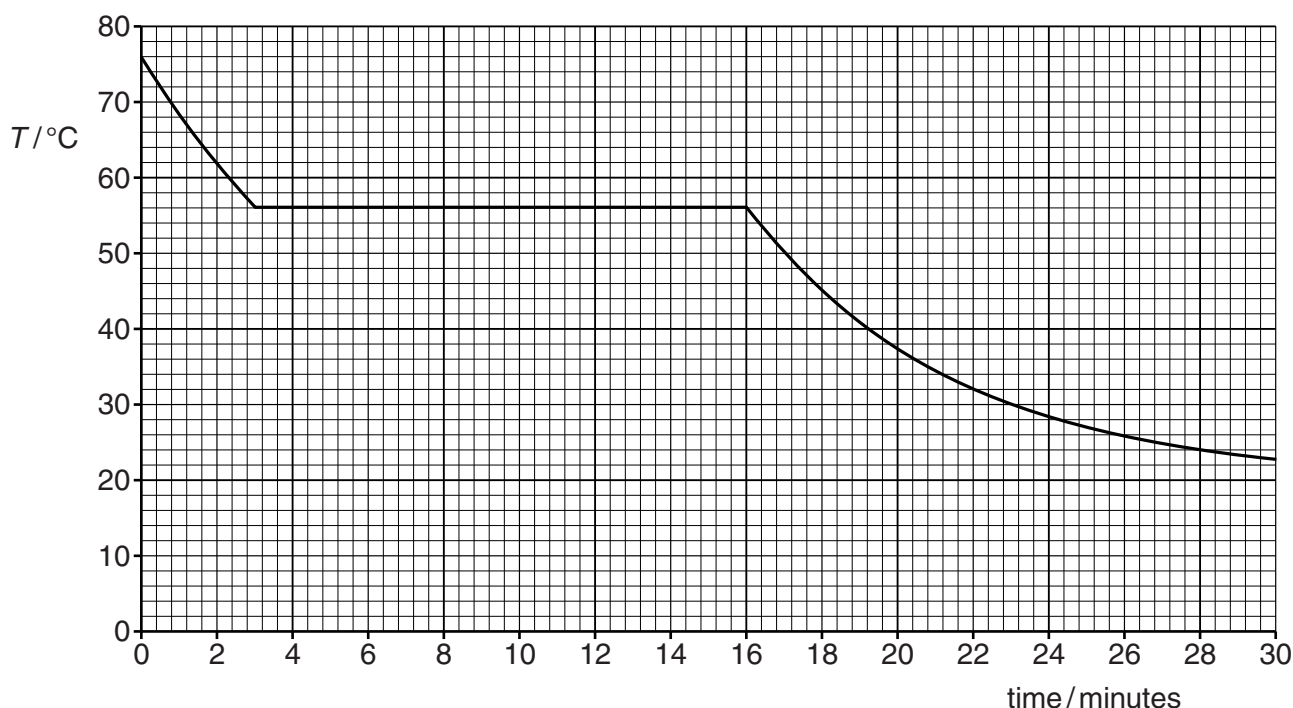


Fig. 10.1

- (a)** Determine the melting point of the wax.

melting point = [1]

- (b)** The test-tube contains 110g of wax that has a specific latent heat of fusion of 210J/g.

Calculate the thermal energy transferred from the wax between 3 and 16 minutes.

energy = [2]

- (c) (i)** State what happens to the wax between 3 and 16 minutes.

..... [1]

- (ii)** Between 3 and 16 minutes, the temperature of the wax is above room temperature and energy is lost to the room.

Explain, in terms of molecules, why the temperature of the wax remains constant.

.....

.....

..... [2]

11 Fig. 11.1 shows apparatus used when determining the specific heat capacity of aluminium.

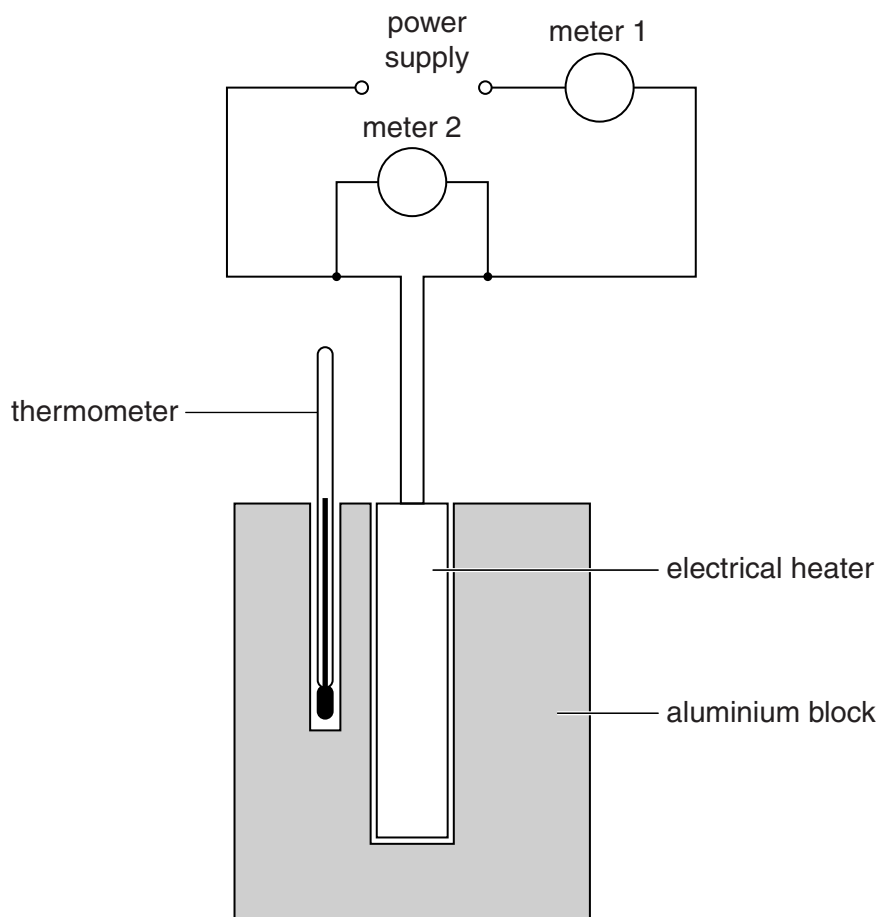


Fig. 11.1

Meter 1 and meter 2 are electrical meters.

(a) State the quantities measured by meter 1 and by meter 2.

meter 1:

meter 2:

[1]

(b) The following results are obtained in the experiment:

initial temperature of block	= 18 °C
final temperature of block	= 40 °C
mass of block	= 0.85 kg
energy supplied to heater	= 17 000 J

Assuming that no heat is lost from the block during the experiment, calculate the specific heat capacity of aluminium.

specific heat capacity = [2]

(c) In practice, some heat is lost from the block and the air above it becomes hot.

(i) Describe and explain the process by which the hot air moves away from the block.

.....
.....
.....
..... [2]

(ii) Suggest how to reduce the loss of heat from the block in the experiment.

.....
..... [1]

[MJ2014/P21/Q2]

12 Most substances expand when they are heated.

(a) (i) State one example where expansion is useful.

Mercury expansion in thermometer [1]

(ii) State one example where expansion causes a problem.

Railway track joints if not spaced may cause accident. [1]

(b) Explain, using ideas about molecules, why solids expand when heated.

On application of heat, molecules of solid vibrate with greater amplitude
now requires more space hence volume increases.

.....
.....
..... [2]

(c) When equal volumes are heated through the same temperature rise, the expansions of solids, liquids and gases are different.

Complete each of the two sentences using one of these expressions:

much larger

slightly larger

much smaller

slightly smaller

1. The expansion of a solid is slightly smaller than the expansion of a liquid.

2. The expansion of a gas is much larger than the expansion of a liquid. [2]

[MJ2014/P22/Q3]

- 13** Two metal saucepans contain the same mass of hot water at the same initial temperature. Pan A is white and pan B is black, but otherwise the two saucepans are identical. Both saucepans are uncovered and cool under the same conditions. The cooling curves for the two saucepans are shown in Fig. 13.1.

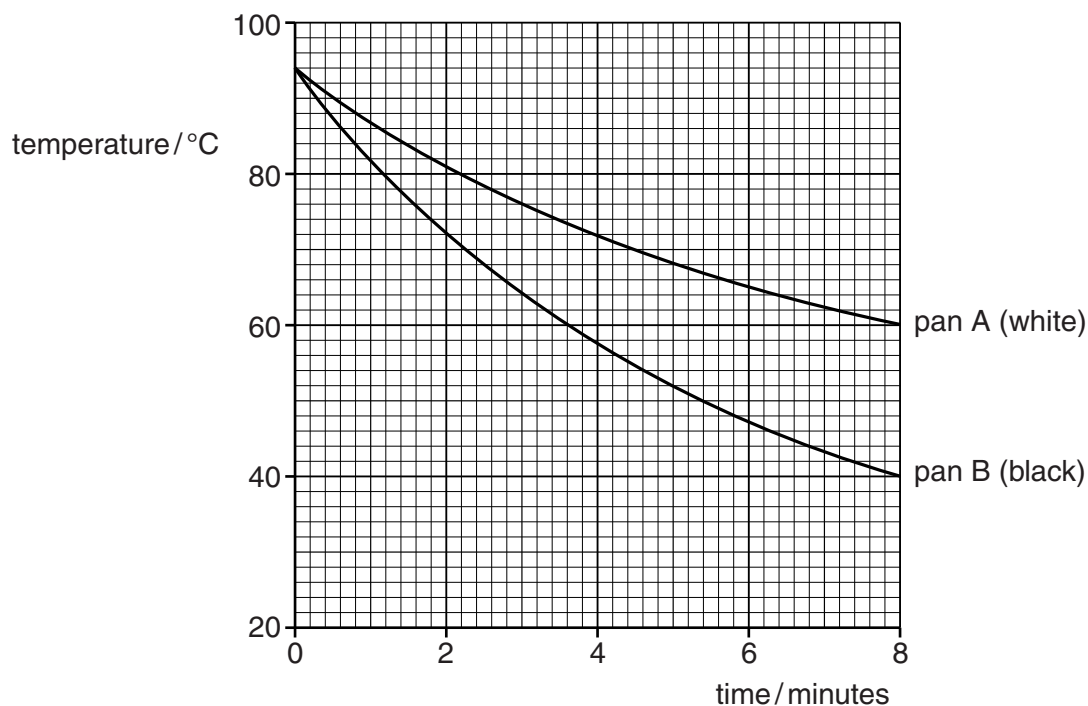


Fig. 13.1

- (a)** Describe how the water in a pan loses heat by

- (i)** conduction,

.....

 [2]

- (ii)** convection.

.....

 [2]

- (b) (i)** Explain why pan B cools faster than pan A.

.....
 [1]

- (ii) Describe and explain how Fig. 13.1 is different when the pans are covered and the experiment is repeated.

.....
.....
..... [2]

- (c) The specific heat capacity of water is $4200 \text{ J/(kg } ^\circ\text{C)}$.

- (i) Explain what is meant by *specific heat capacity*.

.....
.....
..... [2]

- (ii) The specific heat capacity of water is very high. Suggest one disadvantage of this when water is used for cooking.

.....
..... [1]

- (iii) The water in pan A cools for 8 minutes, as shown in Fig. 13.1. During this time, the water loses an average of 9000 J of thermal energy per minute.

1. Calculate the mass of water in pan A.

mass = [3]

2. The mass of water in pan B is the same as that in pan A.

Calculate the thermal energy lost from the water in pan B during the 8 minutes.

loss of thermal energy = [2]

- 14** The casing of an electric kettle is made of white plastic. Fig. 14.1 shows the heating element positioned in the base of the kettle.

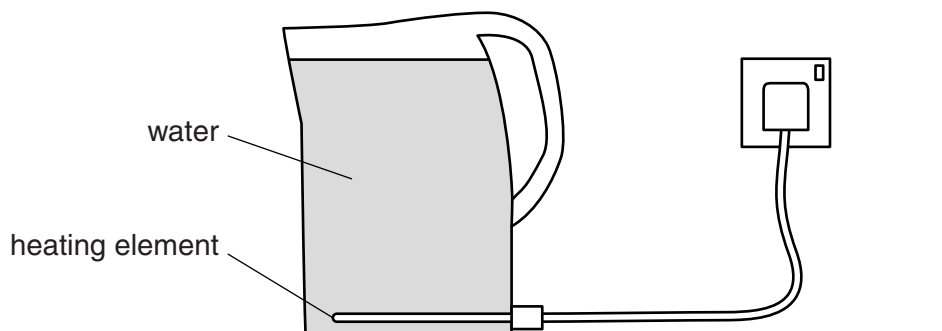


Fig. 14.1

- (a) (i)** The heating element supplies thermal energy to the water at the bottom of the kettle.
Describe and explain how the thermal energy is transferred throughout the water.
.....
.....
.....
.....[3]
- (ii)** Explain why a kettle with its heating element in the water at the top of the kettle does not heat the water uniformly.
.....
.....[1]
- (b)** The kettle is powered by a 230V supply. It is switched on for 3.5 minutes and there is a current of 9.6A in the heating element.
- (i)** Calculate the thermal energy produced in the heating element in this time.

thermal energy =[2]

- (ii) The kettle contains 1.6 kg of water that was at an initial temperature of 22 °C. The specific heat capacity of water is 4200 J/(kg °C).

Calculate the maximum possible temperature of the water.

temperature =[3]

- (iii) Suggest one reason why the temperature of the water, after 3.5 minutes, is less than the value calculated in (b)(ii).

.....
.....[1]

- (c) Explain one advantage of

- (i) using plastic for the casing of a kettle,

.....
.....[1]

- (ii) choosing white as the colour for the outside of the casing.

.....
.....[1]

- (d) The kettle is switched on again and the water reaches its boiling point. It starts to boil and the kettle remains switched on.

- (i) State the meaning of *boiling point*.

.....
.....[1]

- (ii) Explain, in terms of molecules, what happens to the thermal energy that is supplied when the water is boiling.

.....
.....
.....[2]

- 15** A test-tube contains solid wax. The melting point of the wax is 58°C .

The test-tube is partially immersed in a beaker of boiling water and the wax melts. A thermometer is placed in the liquid wax.

At time $t = 0$, the thermometer reads 90°C and the test-tube is immediately removed from the water. The test-tube then cools to room temperature of 23°C .

- (a)** The temperature of the wax changes as it cools to room temperature.

On the axes of Fig. 15.1, sketch a graph to show how the temperature changes with time t . Mark on the temperature axis 90°C , 58°C and 23°C . [3]

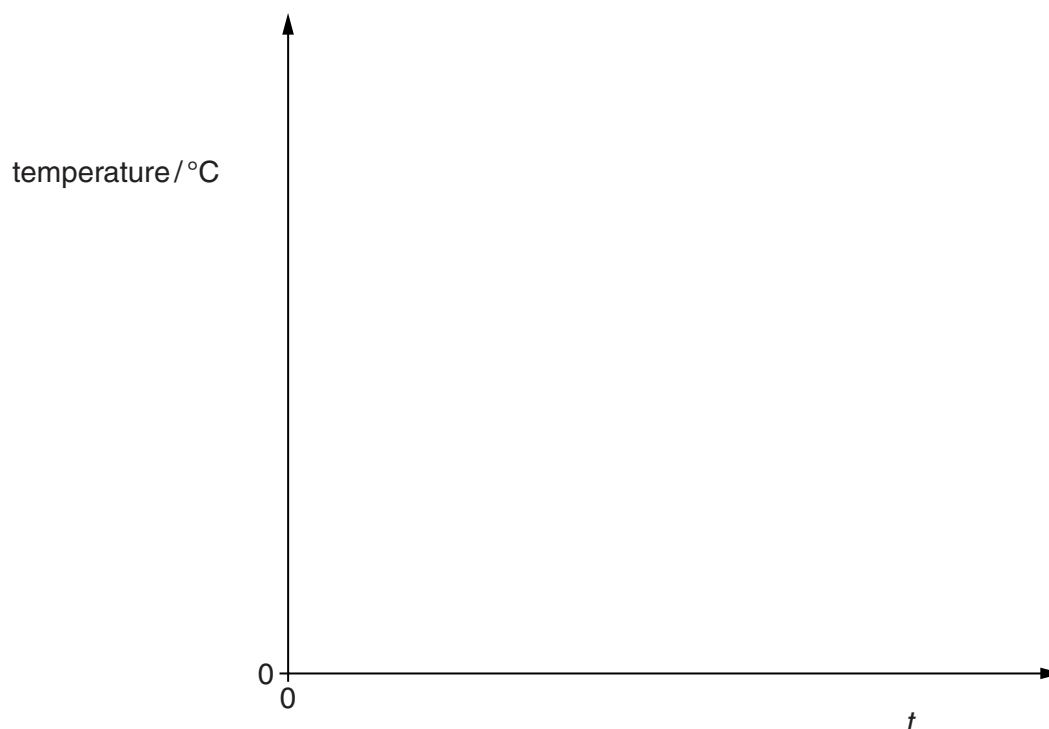


Fig. 15.1

- (b)** On the completed graph of Fig. 15.1, mark a point H, where half of the wax is solid and half is liquid. [1]
- (c)** The specific latent heat of fusion of the wax is 220 J/g and the mass of the wax is 45 g .

Calculate the thermal energy released by the wax as it solidifies.

thermal energy = [2]

[ON2014/P22/Q5]

- 16 (a) State what happens to the molecules of a gas as its temperature increases.

Molecules speeds up & move away from each other.

[1]

- (b) Fig. 16.1 shows a runner in a long distance race. He tips water over himself to keep cool.



Fig. 16.1

- (i) Explain, in terms of molecules, how the evaporation of water keeps the runner cool.

Water on the skin receive heat & molecules at the surface become higher energetic leaves the surface while leaving lesser (colder) energetic molecules behind. This gives cooling effect.

[3]

- (ii) At one point in the race, a strong breeze blows past the runner and the water evaporates more quickly.

Suggest one reason why the water evaporates more quickly in the breeze.

Rate of evaporation is directly proportional to wind speed.

[1]

[ON2014/P22/Q6]

17 (a) A student makes a diagram of the molecules in a liquid, as shown in Fig. 17.1.

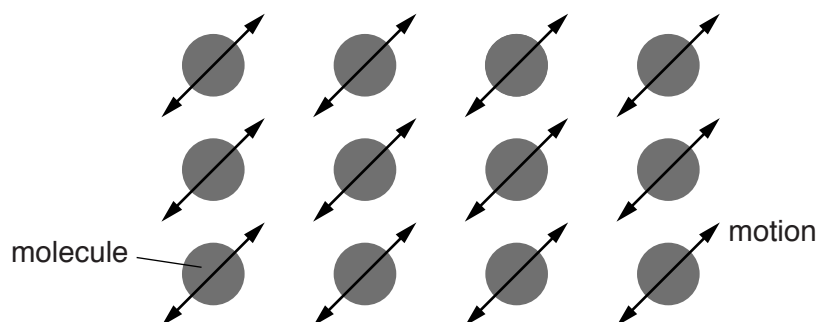


Fig. 17.1

Suggest two ways in which the student's diagram does not show the positions and motion of the molecules accurately.

1.
.....

2.
.....[2]

(b) Some of the molecules of a liquid evaporate.

(i) Describe what happens during evaporation.

.....
.....[1]

(ii) Explain why evaporation has a cooling effect on the liquid.

.....
.....
.....
.....[2]

- (c) The beaker shown in Fig. 17.2 contains pure water at a temperature of 80°C . A thermometer, initially at room temperature, is placed in the air just above the water. The reading on the thermometer starts to increase slowly.

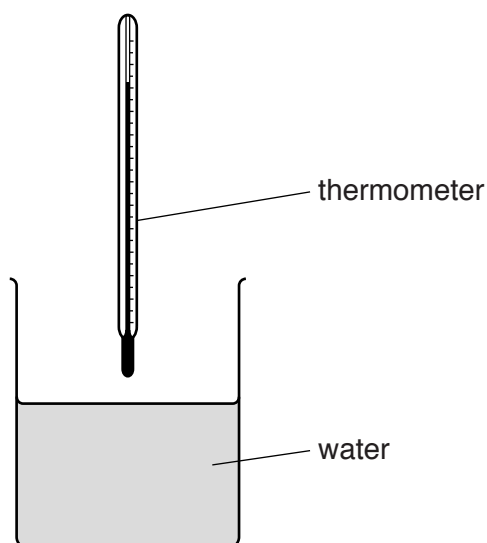


Fig. 17.2

- (i) Explain how convection in the air causes the reading on the thermometer to increase.

.....
.....[1]

- (ii) A heater is placed underneath the beaker and the water boils vigorously to produce steam. The thermometer reading increases quickly.

Explain how the water in the gas state gives thermal energy (heat) to the thermometer.

.....
.....
.....
.....[2]

(iii) The heater placed underneath the beaker has a power of 200W.

The mass of water in the beaker is initially 100g.

The water in the beaker is heated for a time of 120s. During this time, the temperature of the water in the beaker rises from 80°C to 100°C and then the water boils. At the end of this time, the mass of water in the beaker is 95g.

The specific heat capacity of water is 4.2J/(g°C).

The specific latent heat of vaporisation of water is 2250 J/g.

Calculate the thermal energy (heat)

1. produced by the heater in 120s,

thermal energy =[2]

2. used to warm the water from 80°C to 100°C,

thermal energy =[2]

3. used to reduce the mass of water in the beaker,

thermal energy =[2]

4. produced by the heater that is not transferred to the water.

thermal energy =[1]

[MJ2015/P21/Q10]

- 18** A metal can and a plastic bottle, both containing liquid, are cooled by placing them in a jug of melting ice, as shown in Fig. 18.1.

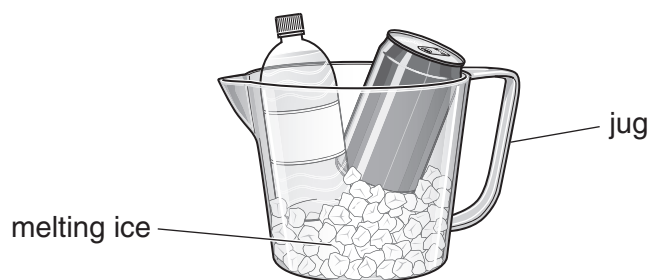


Fig. 18.1

The can and bottle each contain 330 g of the same liquid at 15 °C.

- (a)** The specific heat capacity of the liquid is 4.2 J/(g °C).

Calculate the thermal energy released when 330 g of the liquid at 15 °C cools to 2 °C.

energy = [2]

- (b)** When water at 0 °C is used in the jug, instead of the melting ice, the cooling is slower.

Explain why cooling is faster when using melting ice in the jug, rather than water at 0 °C.

.....
.....
.....
..... [2]

- (c)** The liquid in the metal can cools down faster than the liquid in the plastic bottle.

Suggest why this happens.

.....
.....
..... [1]

19 In a steam cleaner, water is heated to its boiling point and is then turned into steam (water in the gas state).

(a) Define *boiling point*.

.....
..... [1]

(b) (i) Describe the molecular structure of a liquid.

.....
.....
.....
.....
..... [2]

(ii) Explain, using ideas about molecules, why thermal energy (heat) must be supplied to the water as it boils.

.....
.....
.....
.....
..... [2]

[ON2015/P21/Q5]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2011	5054	21

- 4 (a) ($E/Q =$) $mc\Delta T$ in any algebraic or numerical form e.g. 4200×16 , 2100×5 C1
 67 200 or 10 500 or 77 700 seen or ($E/Q =$) ml or mL algebraic seen C1
 4.2×10^5 J A1
- (b) (i) break bonds // separate molecules // give molecules more P.E. B1
- (ii) (different) change in distance // molecules not so far apart // incomplete bond breaking // doesn't push atmosphere back // less work against atmosphere B1 [5]

Page 5	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2011	5054	22

- 11 (a) (i) 50°C and $24/25^\circ\text{C}$ B1
- (ii) heat loss or evaporation mentioned // molecules escape // liquid to vapour C1
more heat loss or **more** evaporation etc. **because** temperature is higher A1
- (iii) temperature becomes 100°C // reaches boiling point // temperature becomes steady B1
 water boils // water turns to steam/gas // energy loss = energy gain B1
- (b) (i) ($c =$) $E/m\Delta T$ in any form, numerical or algebraic, e.g. $7400/72 \times 23$ C1
 4.5 or 4.47 or 4.4686 J/(g $^\circ\text{C}$) // 4468.6 J/(kg $^\circ\text{C}$) A1
- (ii) ($E =$) $\frac{1}{2}mv^2$ algebraic only C1
 $\frac{1}{2} \cdot 0.072 \cdot 450^2$ C1
 7300 J // 7290 J (7 290 000 (J) alone gets 2/3) A1
- (iii) *water molecules* move/vibrate fast(er)/(more) vigorously B1
water molecules **random** motion // move (more) throughout liquid/all directions // slide over each other // move in convection // hit more often // move further apart B1
bullet molecules motion in one direction/away from gun/towards target/ all have same (increase in) speed B1
- (c) two different metals at any junction/two outside wires if three used B1
 joined wires connected to meter/voltmeter/ammeter/galvanometer B1 [15]

Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – October/November 2011	5054	21

- 1 (a) (i) ($V =$) 64 or 6.4×10^{-5} or 4^3 or 0.04^3 C1
 ($m =$) ρV or $920 \times 6.4 \times 10^{-5}$ or 920×0.04^3 C1
 0.059 kg or 59 g or 0.05888 kg A1
- (ii) 0.59 N or 0.5888 N B1
- (b) ($Q =$) ml or $0.059 \times 3.4 \times 10^5$ C1
 $2.0(0) \times 10^4$ / $2.0(1) \times 10^4$ / 2.006×10^4 J A1 [6]

Page 4	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – October/November 2011	5054	22

- 10 (a) **temperature** where: liquid and solid may exist together **or** solid turns to liquid B1
- (b) (i) $(E =) ml$ C1
 $0.0019 \times 2.2 \times 10^4$ **or** $1.9 \times 2.2 \times 10^4$ **or** 41 800 **or** 42 000 C1
42 (41.8) J A1
- (ii) $\frac{1}{2}mv^2$ **or** $\frac{1}{2} \times 0.0019 \times v^2$ **or** $\frac{1}{2} \times 1.9 \times v^2$ C1
 $(v^2 =)$ 44 000 **or** 44 C1
210 (209.761 etc.) m/s A1
- (iii) any **two** of:
heat lost to wall
heat to raise bullet to m.p.
air resistance/air friction reduces energy/speed/velocity **or** work done
against **air resistance/air friction** (in air/as bullet travels) B2
- (c) any **three** of:
molecules become further apart
molecules become randomly positioned/less ordered
molecules moving throughout liquid/in clusters/were fixed/free to move/
slide over each other
bonds broken/overcome/weaker **or** forces reduced B3
- (d) twice the energy needed **OR** $ml = \frac{1}{2}mv^2$ M1
(bullets have) twice the KE m cancels **or** mass irrelevant **or** w.t.t.e.
or calculation M1
they melt A1 [15]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2012	5054	21

- 4 (a) **temperature** when solid melts;
temperature when solid changes to liquid B1
- (b) (i) temperature increases; molecules move faster/have more kinetic energy/
vibrate faster/move further apart B1
- (ii) change of state; solid changes to liquid B1
latent heat provided; break bonds; molecules move apart/break free;
reduce bond strength; idea of more disorder B1
- (c) liquids expand more than solids B1 [5]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2012	5054	22

- 4 (a) 80 °C B1
- (b) $(Q =) mcT$ in any form numerical or algebraic C1
1530 J A1
- (c) (i) Any 2 lines from
• **latent** heat/energy mentioned
• **latent** heat/energy given out/lost
• bonds being made/strengthened
• molecules lose PE
• molecules KE constant B2
- (ii) ANY 2 lines but max 1 if no change/comparison implied
molecules change **from** OR in liquid
• random arrangement
• move throughout in some form (e.g. move freely)
• move or occur in clusters
change **to** OR in solid
• regular arrangement/shape or fixed position/shape
• vibrate
• separation (probably) close(r) B2 [7]

Page 5	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2012	5054	21
10 (a)	16 × 7.5 or 120 or 96–17 or 79 (Q =) $mc\Delta T$ or 120 × 2300 × 79 2.2(2.1804) × 10 ⁷ J	C1 C1 A1	[3]
(b) (i)	2.2 × 10 ⁷ /7 or 2.2 × 10 ⁷ / (7 × 60) or 2.2 × 10 ⁷ / (7 × 3600) 3.1 × 10 ⁶ J/h or 5.2 × 10 ⁴ J/min or 870 J/s or W	C1 A1	
(ii)	(heater/bricks) hot(ter) (not room cooler) great(er) temperature difference (between heater and room)	B1 B1	[4]
(c)	air (next to heater) gets hot or conduction through metal/casing expands or radiation or IR (radiation) less dense rises circulation or convection current or arrows on Fig. 10.2	B1 B1 B1 B1 B1	[5]
(d)	double glazing/cavity walls/ceiling tiles/carpet/curtains/loft insulation etc. or shiny foil traps air radiation reflected air is poor conductor/convection IR radiation/ prevented back into room	B1 M1 A1	[3]

Page 4	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – May/June 2013	5054	21
10 (a) (i)	(amount of) energy/work (by a device of power) 1 kW in 1 hr	M1 A1	
(ii)	80/1000 or 0.08 seen (e.g. 0.08 × 24 × 25) 168 or 24 × 7 (hours) seen (e.g. 0.08 × 24 × 7 × 25) 336 c or 340 c (accept \$3.36 or any other e.g. £, R)	C1 C1 A1	[5]
(b) (i)	mcT algebraic or numerical conversion of mass to g seen, e.g. 1500 used or shc used as 4200 1.6 × 10 ⁵ J or 1.58 × 10 ⁵ J or 157 500 J (allow 157(.5) J to score 2/3)	C1 C1 A1	
(ii)	($m=$) E/L in any form numerical or algebraic e.g. 157 500/3.3 × 10 ⁵ 0.48 or 0.477 kg e.c.f. (i)	C1 A1	
(iii)	1. no fixed position/clusters/arranged randomly/close together/closely packed move throughout/at random/slide past each other/not in an organised way 2. regular/orderly arrangement/crystal lattice or fixed position or close together (ign. evenly spaced) vibrate	B1 B1 B1 B1	
(iv)	nothing/no change and increases	B1	[10]

Page 2	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2013	5054	21
1 (a)	($m =$) ρV or 1000 × 450 4.5 × 10 ⁵ kg	C1 A1	
(b) (i)	(Q =) $mc\Delta T$ or 4.5 × 10 ⁵ × 4.2 × 15 or 4200 and 15/(27–12) 4.5 × 10 ⁵ × 4200 × 15 or 2.8(35) × 10 ⁷ 2.8(35) × 10 ¹⁰ J	C1 C1 A1	
(ii)	thermal/internal energy/heat lost or gained by something specific (e.g. air/pool walls/tiles etc.) or heat lost by evaporation	B1	[6]

Page 3	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2013	5054	22

- 4 (a) 56 °C (**not** ° or C°) B1
 (b) (Q =) *ml* **or** 110 × 210 C1
 2.3(1) × 10⁴ J A1
 (c) (i) (wax) is solidifying **or** freezing B1
 (ii) (molecules) form structure/come closer/lose PE M1
or bonds made/stronger (no e.c.f. from (c)(i))
 K.E. of molecules const. **or** replace/release/produce energy/heat
 (transferred to environment/latent heat emitted)
 (no e.c.f. from (c)(i)) A1 [6]

Page 2	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – May/June 2014	5054	21

- 2 (a) current **and** voltage / p.d. / e.m.f. in correct order [B1]
 (b) (c=) E / mT in symbols or numbers e.g. 17 000 = 0.85 × c × 22 [C1]
 910 J / (kg °C) [A1]
 (c) (i) (hot air) rises **or** convection mentioned [B1]
 (hot) air less dense [B1]
 (ii) lag **or** cover with insulating material **or** warmer room **or** start with colder block [B1]

Page 2	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – May/June 2014	5054	22

- 3 (a) (i) any sensible example where expansion is useful [B1]
 (ii) any sensible example where expansion causes a problem [B1]
 (b) (molecules) move fast(er) or vibrate fast(er) or have more (kinetic / potential / internal) energy [B1]
 (molecules) move apart or distance between molecules increases or vibration has larger amplitude or vibration takes up more space or bonds stretch [B1]
 (c) slightly smaller [B1]
 much larger [B1]

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2014	5054	21

- 10 (a) (i) **heated / hot water** expands **or** density of **heated / hot water** decreases B1
 (heated / hot water) rises B1
 convection (current) / circulation set up **or** (heated / hot water) rises **and** cold water sinks B1
 (ii) convection transfers heat upwards **or** less dense / heated / hot water (already) at top B1 [4]
 (b) (i) (Q =) *VIt* **or** 230 × 9.6 × 3.5 **or** 230 × 9.6 × 3.5 × 60 **or** 7728 C1
 4.6(368) × 10⁵ J A1
 (ii) (ΔT =) *Q / mc* **or** 4.6(3680) × 10⁵ / 1.6 × 4200 C1
 69 (°C) C1
 91 °C A1
 (iii) evaporation **or** thermal energy / heat in **plastic casing / element / surroundings** (i.e. air or environment) B1 [6]
 (c) (i) poor conductor (heat or electricity) **or** less heat lost / cooler to touch **or** less risk of shock B1
 (ii) poor emitter **and** less heat lost / of radiation / IR (**not** poor absorber) B1 [2]
 (d) (i) **temperature** where liquid and vapour / gas coexist **or** where liquid (**not** substance) boils (at atmospheric pressure) (**allow** becomes vapour / gas) B1
 (ii) (work done) against / overcoming forces between molecules **or** molecules gain P.E. (**ignore** K.E. increases) B1
 changes to P.E. / molecules separate B1 [3]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2014	5054	22
5	(a) downward curve of correct curvature from marked 90 °C horizontal line at marked 58 °C downward (asymptotic) curve of correct curvature to marked 23 °C (b) H marked halfway (by eye) along an intermediate horizontal line (c) (Q =) mL or 45×220 9900 J	B1 B1 B1 B1 C1 A1	[6]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2014	5054	22
6	(a) (the molecules) move faster or have more kinetic energy or accelerate ignore vibrate faster (b) (i) faster/energetic molecules escape average speed decreases or slower molecules remain temperature depends on average KE or heat taken from runner OR liquid becomes gas/vapour latent heat needed or bonds broken heat taken from runner (ii) water vapour blown away or surrounding air less humid	B1 B1 B1 B1 B1	[5]

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	21
10	(a) (liquid) molecules not arranged (so) regularly (liquid) molecules not vibrating/moving in same direction or do not have same speed (b) (i) molecules/liquid escape (from surface)/break bonds (ii) fast moving/more energetic molecules evaporate/escape leaving slow molecules or molecules with less kinetic energy (on average) (c) (i) hot air rises (ii) (steam) condenses or changes to liquid (on thermometer) or heat (conducted) from hot to cold gives out latent heat (to thermometer) or explanation involving bonds being made (iii) 1 (E =) Pt or 200×120 24 000 J 2 (E =) mcT or $100 \times 4.2 \times 20$ 8400 J 3 (E =) mL or 5×2250 11 250 J 4 4350 J or 1 – (2+3)	B1 B1 B1 B1 B1 B1 B1 B1 C1 A1 C1 A1 C1 A1 B1	

Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	22
4	(a) (H=) mcT or $330 \times 4.2 \times 13$ 18 000 J or 18 020 J or 18 018 J (b) ice takes in/needs heat/energy • for latent heat • to melt/turn to water (at 0 °C)/change state • to break bonds/for molecules to gain P.E. water (in jug initially at 0 °C) warms up or ice (and melted water in jug) • stays at 0 °C/stays cold/stays at constant temp. • gives larger temperature difference (between liquid and melting ice in jug) (c) metal is a good conductor (of heat) or metal/can has lower heat capacity allow opposite statements for plastic, e.g. plastic is an insulator (of heat), penalise wrong statements and Physics, e.g. liquid evaporates from can, metals conduct temperature/convect better	C1 A1 B1 B1 B1	

Page 3	Mark Scheme	Syllabus	Paper
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- 5 (a) **temperature** at which a liquid becomes a gas B1
- (b) (i) molecules close together/ touching **or** closer than in gas B1
randomly arranged **or** irregular structure B1
- (ii) to separate/ increase the distance between molecules B1
work done against (intermolecular) forces **or** supply p.e. **or** break bonds B1 [5]